

Final Project: 3-D Reconstruction from a Slit-Laser Turntable Scanner via Triangulation

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1 Problem and Coordinate Frame

A green slit laser projects a vertical plane that passes through the rotation axis of a hexagonal turntable. A fixed camera captures the scene while the table (and the gnome on it) rotate in 2° steps, giving 180 frames over 360° in two sets: `withLaser` and `withoutLaser` (720×1280). The goal is one coloured 3-D point cloud.

World frame. Origin at the turntable centre, $+Z$ = rotation axis (up). From `GroundTruth.ply` the turntable is a regular hexagonal prism of circum-radius $R = 25$ mm (vertex–vertex diameter 50 mm), top face $z = +1$, bottom $z = -14$ (thickness 15 mm). Reconstruction is performed in this frame; `GroundTruth.ply` is used *only* for the final error analysis.

The pipeline is the classical laser-plane triangulation: each laser pixel back-projects to a camera ray which is intersected with the (calibrated) laser plane to give a 3-D point; points are then de-rotated by the frame angle and accumulated.

2 Camera Calibration (Extrinsics and Focal)

A world point $\mathbf{P}$ projects as $s[u, v, 1]^\top = K[R | \mathbf{t}][\mathbf{P}, 1]^\top$. We recover $\{R, \mathbf{t}\}$ from the turntable, whose metric geometry is known. In `withoutLaser/0000.jpg` the prism silhouette is segmented (bright, low saturation) and six corners are located sub-pixel by intersecting line-fits of the silhouette edges (Table 1). These are matched to the hexagon vertices (one near vertex with both side vertical edges visible $\Rightarrow$ three consecutive vertices) and the pose is solved with `cv2.solvePnP`.

Focal correction. The nominally provided $f = 720$ px is *not* consistent with the imaged 50 mm turntable (best reprojection > 22 px at any pose). Refining f together with the pose gives $f \approx 1281$ px ($= 720 \times 16/9$, a render FOV/sensor-fit artefact) and a reprojection RMS of **0.04 px**; the principal point $(c_x, c_y) = (360, 640)$ is the image centre and checks out exactly. We therefore adopt

$$K = \begin{bmatrix} 1281 & 0 & 360 \\ 0 & 1281 & 640 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{C} = -R^\top \mathbf{t} = (-112.6, 10.8, 25.4) \text{ mm},$$

i.e. camera distance 116 mm, azimuth 174.5° , elevation 12.7° . The world-frame azimuth is a free gauge (fixed by placing one vertex at 90°); the table's true frame-0 azimuth becomes a global yaw vs. the ground truth and is irrelevant to shape.

Table 1: Six turntable corners on frame 0 and per-corner reprojection error ($f=1281$). Mean RMS = 0.04 px, max = 0.10 px.

corner	A _{top}	A _{bot}	B _{near}	C _{bot}	C _{top}	D _{top}
(u, v) px	(84,921)	(84,1095)	(224,1198)	(572,1184)	(572,977)	(648,911)
err (px)	0.10	0.07	0.03	0.03	0.02	0.01

3 Laser-Plane Estimation

Because the beam passes through the rotation axis, the laser plane contains the z -axis: $ax + by = 0$. We isolate the stripe that lands on the *bare* turntable top face ($z = +1$) over the first frames (inside the projected top hexagon, on the bright table only), back-project each pixel to $z = +1$, and fit a line through the origin. From 948 points:

$$-0.504x - 0.864y = 0 \quad (\text{residual RMS} = 0.30 \text{ mm}).$$

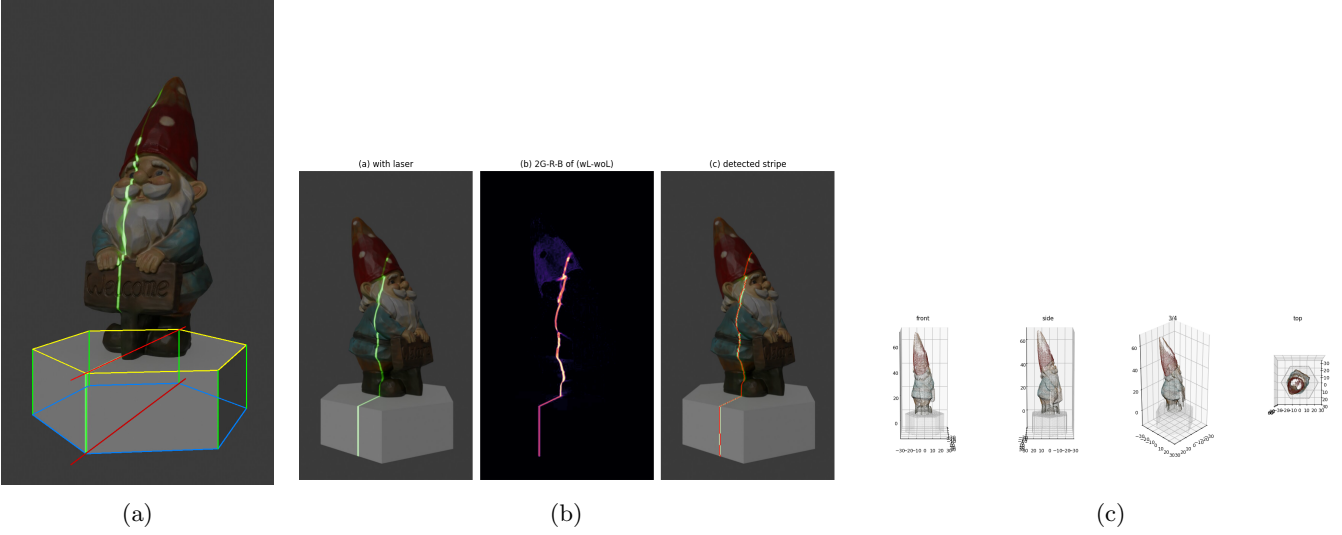


Figure 1: (a) Calibration overlay: reprojected prism (yellow/blue/green) and laser plane (red) on the scan. (b) Stripe detection: with-laser, $2G - R - B$ response, and detected sub-pixel stripe (red). (c) Reconstructed coloured cloud (42,341 pts) – front, side, 3/4, top.

Figure 1a overlays the reprojected prism wireframe and this plane on the scan, confirming the calibration.

4 Stripe Detection and Triangulation

Detection. Per frame, the excess-green response $\text{resp} = 2G - R - B$ of (withLaser – withoutLaser) isolates the stripe (Fig. 1b). The green laser is largely absorbed by the *red* hat, so $G - \max(R, B)$ detects only $\sim 50\%$ of the hat rows (a hole at the tip); the excess-green index is far more sensitive there ($\sim 64\%$ at the same threshold) while keeping a 0% background false-positive rate. Per image row the stripe column is the intensity-weighted centroid around the peak (sub-pixel, ± 5 px window, threshold 35).

Triangulation. For pixel (u, v) the world ray is $\mathbf{d} = R^T K^{-1}[u, v, 1]^T$ from $\mathbf{C}$; intersecting $ax + by = 0$ gives $s = -(aC_x + bC_y)/(ad_x + bd_y)$ and $\mathbf{X} = \mathbf{C} + s\mathbf{d}$. Colour is bilinearly sampled from *withoutLaser* (no green tint).

Rotation direction (image-only). Tracking the near-vertex azimuth shows it increases $+2^\circ/\text{frame}$, so the table turns CCW about $+z$; a point seen at frame i is de-rotated by $R_z(-2^\circ i)$ into the frame-0 object frame and accumulated. The cloud is then cropped to the scan volume ($r \leq 26$ mm), down-sampled to a 0.4 mm voxel grid, and cleaned by a k -NN distance test plus a radius-outlier test (which removes the faint laser spray above the hat tip). Result: $160,037 \rightarrow 42,341$ coloured points (Fig. 1c).

5 Results and Error Analysis

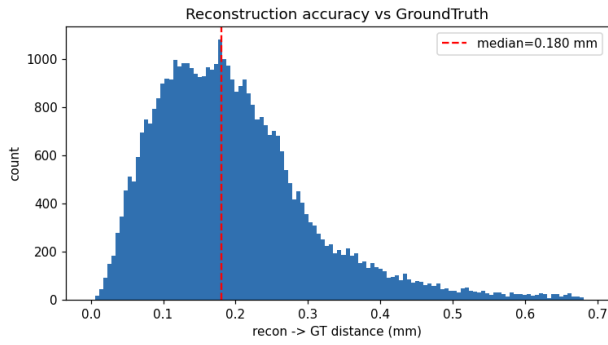
The cloud is compared to a dense (4×10^5 pts) area-weighted sampling of the GT mesh. Sharing origin, scale and z -axis with the GT, the only free DOF is a yaw; a 1° yaw search (300° , no mirror $\Rightarrow$ handedness correct) plus a few rigid-ICP steps aligns them (this alignment is an evaluation-only diagnostic). Distances are reported both ways (Table 2, Figs. 2a–2b).

Table 2: Cloud-to-cloud distances (mm). recon $\rightarrow$ GT = accuracy; GT $\rightarrow$ recon = coverage.

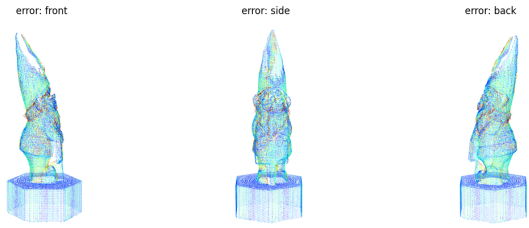
direction	mean	median	RMS	95%	max
recon $\rightarrow$ GT (accuracy)	0.203	0.180	0.244	0.424	2.55
GT $\rightarrow$ recon (coverage)	1.48	0.30	3.41	9.09	16.9

Accuracy is excellent: median 0.18 mm and 95% below 0.44 mm, i.e. well under 1% of the object size, consistent with the 0.04 px calibration and 0.30 mm plane fit. The error map shows the largest residuals on steep/oblique surfaces (hat sides, beard, sign edges) where the stripe centroid is least precise.

Coverage. 78.3% of the GT lies within 1 mm of a reconstructed point. The uncovered $\sim 22\%$ (long GT $\rightarrow$ recon tail) are regions a single fixed camera+laser cannot see: deep concavities, surfaces between the legs and under the welcome sign/arms, and the turntable underside. These are intrinsic occlusions of the setup, not calibration error – accuracy where the laser *is* seen remains sub-millimetre.



(a)



(b)

Figure 2: (a) Histogram of recon→GT distance. (b) Error map (jet: blue 0 mm → red ≥ 0.5 mm); most of the surface is sub-0.2 mm, with larger error only at steep edges/fine detail.

Conclusion. A fully image-derived calibration (PnP on the known turntable, focal corrected to 1281 px, z -axis laser plane) plus background- subtraction stripe detection and ray-plane triangulation reconstructs the gnome to a median accuracy of 0.18 mm. Deliverables: `calibrate.py`, `reconstruct.py`, `error_map.py`, and `M11415015.xyz`.